

C Language Learning Roadmap 🚀🔧

This roadmap is designed to take you from a C language beginner to an advanced level. It outlines the key concepts and topics you should grasp to become proficient in C programming. Follow this roadmap step by step, practice, and build projects to reinforce your learning. 📚💪

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Introduction 🌱

- What is C language and its history
- Setting up a C development environment
- Writing and compiling a basic C program

****Mini Project Task Challenge****: Create a simple "Hello, World!" program in C.

Basic Syntax 🖋️

- Structure of a C program
- Comments and whitespace
- Writing and using functions

****Mini Project Task Challenge****: Develop a C program to find the factorial of a given number.

Data Types

- Basic data types (int, float, char, etc.)
- Modifiers (signed, unsigned, short, long)
- Type conversions and casting

****Mini Project Task Challenge****: Write a program to convert temperatures between Fahrenheit and Celsius.

Variables and Constants

- Declaring and initializing variables
- Using constants and literals
- Scope and lifetime of variables

****Mini Project Task Challenge****: Create a program to calculate the area of a circle with a given radius.

Operators

- Arithmetic, relational, and logical operators
- Bitwise and assignment operators
- Ternary operator (conditional operator)

****Mini Project Task Challenge****: Build a program to perform basic calculator operations.

Control Statements

- If-else statements
- Switch-case statements
- Loops (for, while, do-while)

****Mini Project Task Challenge****: Develop a program to check whether a number is prime or not.

Arrays and Strings 🎵

- Declaring and using arrays
- Manipulating strings and string functions

****Mini Project Task Challenge****: Implement a program to reverse a given string.

Functions 📋

- Creating and calling functions
- Passing arguments to functions
- Recursive functions

****Mini Project Task Challenge****: Write a program to find the factorial of a number using recursion.

Pointers 🔍

- Understanding pointers and memory addresses
- Pointer arithmetic
- Using pointers with arrays and functions

****Mini Project Task Challenge****: Create a program to swap two numbers using pointers.

Structures and Unions 🏛️

- Creating and using structures
- Nested structures and arrays of structures
- Unions and their differences with structures

****Mini Project Task Challenge****: Build a program to store and display information about books using structures.

File Handling 📁

- Reading from and writing to files
- Working with file streams

****Mini Project Task Challenge****: Develop a program to read data from a file and display its contents.

Dynamic Memory Allocation

- Allocating memory dynamically with malloc and free
- Creating and using dynamic arrays

****Mini Project Task Challenge****: Write a program to implement a dynamic list using pointers.

Preprocessor Directives

- Working with preprocessor directives
- Using conditional compilation

****Mini Project Task Challenge****: Implement a program with conditional compilation to display system information.

Error Handling

- Handling errors in C programs
- Using errno and perror for error reporting

****Mini Project Task Challenge****: Develop a program that handles division by zero gracefully.

Mini Project Tasks

1. ****Student Record Management****: Create a program to manage student records (name, roll number, marks, etc.).
2. ****Simple Text Editor****: Build a basic text editor with file read, write, and search functionalities.
3. ****Banking System****: Develop a simple banking system to perform basic transactions.
4. ****Employee Payroll****: Create a program to manage employee payroll details and calculate salaries.

5. **Library Management**: Implement a library management system to handle book borrowing and returning.

Remember, the key to mastering C programming is practice and building real projects. Start with the basics, gradually move to more complex topics, and challenge yourself with mini projects along the way. Happy coding!





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